18

Modern methods of texture measurement

D. Kilcast, Leatherhead Food Research Association

18.1 Introduction: texture and food quality

What drives us to eat the food that forms our chosen diet? In prosperous societies, we have available an enormous and ever-increasing range of foods, and manufacturers find themselves in an intensely competitive situation. In less well-developed societies, hunger will be the constant driving force, and our diet will be determined by availability of any food that satisfies our basic nutritional needs.

The consequence of examining these extreme circumstances is that we must recognise that, in trying to understand what consumers desire from food, no single factor can be considered in isolation from other factors. For some years, psychology researchers have been developing models to understand consumer behaviour (e.g. Shepherd and Sparks, 1994). Although there are many possible circumstances under which non-sensory factors such as price and nutritional image can have dominant effects, the sensory characteristics of foods are central to continued purchase of foods.

The sensory evaluation of food is frequently defined by the term ‘tasting’, but this is clearly inadequate in describing all the perceptions involved in eating food. During the sequences of actions that constitute food consumption we perceive a whole range of different characteristics relating to the appearance, flavour and texture of the food. Numerous tools are available for investigating the sensory properties of foods, and the information required must be carefully defined if appropriate tools are to be selected. Systematic development of new products will inevitably depend on the use of different tools at different stages of the development cycle.

18.1.1 The human senses

It is generally accepted that human beings have five senses in operation, namely sight, smell, taste, touch and hearing, although warmth, cold, movement and pain may also be considered as senses of importance in a food context (Fig. 18.1). Foods are complex mixtures of organic and inorganic compounds, arranged into structural units. The perception of the sensory characteristics of foods results from the stimulation of all our

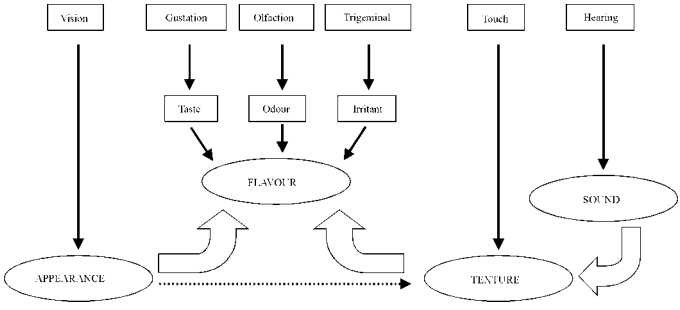


Fig. 18.1 Schematic diagram of the human senses and their operation in the perception of food quality.

senses to some extent by the physicochemical properties of the foods (see also Chapter 2, Section 2.2). The sensory characteristics of food are generally grouped into three categories, namely appearance, flavour and texture. These categories are, however, not independent of one another. For example, colour, which is obviously an important appearance characteristic, can be shown to have an influence on flavour perception; consumers will assign higher scores for flavour intensity to darker foods than to lighter foods. The interaction between appearance and flavour is referred to as 'visual flavour'. Similarly, textural characteristics such as viscosity can influence the perception of flavour, and some flavour characteristics, e.g. acidity, can affect textural characteristics. One means of defining flavour, texture and appearance is by taking into account the fact that each can be attributed to the stimulation of one or possibly two of the senses. On this basis the International Standards Organisation (ISO, 1992) has proposed working definitions for flavour, texture and appearance, as given below.

- *Appearance*: sensory characteristics of foods perceived largely by way of the visual sense. Input from other senses, especially smell, may contribute.
- *Flavour*: the combination of taste and odour. Pain, heat, cold, tactile and visual sensations may also contribute.
- *Texture*: sensory characteristics perceived largely by way of the senses of movement and touch. Input from other senses, especially vision and taste, may sometimes contribute.

The above definitions give little information on how the senses are used in the perception of quality attributes. Appearance is sometimes, mistakenly, equated only with colour, and yet many other visual aspects of form, shape, translucency, etc., may influence our use of the visual senses. Taste (gustation) is strictly defined as the response by the tongue to soluble, involatile materials. These have classically been defined as four primary basic taste sensations: salt, sweet, sour and bitter, although umami, the sensation associated with monosodium glutamate is now becoming recognised as a basic taste, and in some countries this list is extended further to include sensations such as metallic and astringency. The taste receptors are organised groups of cells, known as taste buds, located within specialised structures called papillae. These are located mainly on the tip, sides and rear upper surface of the tongue. Sweetness is detected primarily on the tip of the tongue, salt and sour on the sides of the tongue and bitter on the rear of the tongue. Taste stimuli are characterised by the relatively narrow range between the weakest and the strongest stimulants (*c.* 10^4), and are strongly influenced by factors such as temperature and pH (Meilgaard *et al.*, 1999).

The odour response is much more complex, and odours are detected as volatiles entering the nasal passage, either directly via the nose or indirectly through the retronasal path via the mouth. The odorants are sensed by the olfactory epithelium, which is located in the roof of the nasal cavity. Some 150–200 odour qualities have been recognised, and there is a very wide range (*c.* 10^{12}) between the weakest and the strongest stimulants (Meilgaard *et al*., 1999). The odour receptors are easily saturated, and specific anosmia (blindness to specific odours) is common. It is thought that the wide range of possible odour responses contributes to variety in flavour perception. Both taste and odour stimuli can be detected only if they are released effectively from the food matrix during the course of mastication.

The chemical sense corresponds to a pain response through stimulation of the trigeminal nerve. This is produced by chemical irritants such as ginger and capsaicin (from chilli), both of which give a heat response, and chemicals such as menthol and sorbitol, which give a cooling response. With the exception of capsaicin, these stimulants are characterised by high thresholds. The combined effect of the taste, odour and chemical responses gives rise to the sensation generally perceived as flavour, although these terms are often used loosely.

Texture is perceived by the sense of touch, and comprises two components: somesthesis, a tactile, surface response from skin, and kinesthesis (or proprioception), which is a deep response from muscles and tendons. For many foods, visual stimuli will generate an expectation of textural properties. The touch stimuli themselves can arise from tactile manipulation of the food with the hands and fingers, either directly or through the intermediary of utensils such as a knife or spoon. Oral contact with food can occur through the lips, tongue, palate and teeth, all of which provide textural information.

18.1.2 Texture and food enjoyment

Most studies which have investigated the importance of different sensory modalities on consumer acceptability conclude that flavour is the most important modality, followed by texture and then appearance (e.g. Moskowitz and Krieger, 1995). Such conclusions do not reflect the enormous efforts that the food industry devotes to designing appealing textural characteristics, and in maintaining those characteristics in long-term production. Research with consumers in the United States carried out by Szczesniak and Kahn (1971) showed that awareness of texture lies at a subconscious level, and that textural properties are taken for granted. If the expectations of texture are violated, however, awareness of textural defects are accentuated, and texture becomes a focal point for criticism and rejection of the food. The importance of expectations are being increasingly recognised as important factors in food choice by consumers (e.g. Vickers, 1991; Cardello, 1994).

18.1.3 Importance of texture control and design

Texture and food structure are inextricably linked; the micro- and macro-structural composition of foods will determine the sensory perception, and any change in structure carries the risk of changing perceived texture and violating consumer expectations. Industry therefore needs to take great care to ensure that products with key textural characteristics, such as snack foods and confectionery products, are manufactured to consistently high quality. This can present an enormous challenge for foods relying on primary components such as meat and vegetables that are naturally subject to high

variability, and for any processed food manufactured on high-volume production lines. Product modifications, for example to produce low-fat variants, can introduce structural changes that can generate substantial textural modifications. Industry therefore needs methods to measure textural characteristics. However, designing suitable measurement systems requires an understanding of the mechanisms by which texture is perceived.

18.2 Perception of food texture

18.2.1 Oral breakdown processes

The importance of the interaction between the texture of foods and their perceived flavour can be clearly seen if the time course of events during food consumption is considered. A strong expectation of the flavour and texture characteristics can be generated before the food is introduced into the mouth. As food enters the mouth, and is either bitten or manipulated between tongue and palate, catastrophic changes occur to the structure of the food that strongly influence the way in which tastants and odorants are released from the food. Of particular importance are temperature increase (cold foods) or decrease (hot foods) and dilution by saliva. Salivary introduction also serves to lubricate the food bolus. The factors that influence such release are under active study (e.g. Overbosch *et al.*, 1991), and include:

- rate and mode of production of new surface
- rate of production of saliva
- dissolution and dispersion of the food
- release of involatile tastants and volatile odorants
- transport of volatiles to the nasal cavity

18.2.2 Oral food management

Even the complex picture of food breakdown described previously is an over-simplification of actual oral processes (Heath and Prinz, 1999), which show substantial differences for different foods and between individuals. The first bite by the incisors is an important stage which generates an early textural response that can influence subsequent chewing actions. The food is then transported between the cheek teeth, the jaw closes and main-sequence mastication starts. Hard foods are comminuted into particles, which are then formed into a soft bolus with saliva. Before this bolus is swallowed, the tongue is used to clear any remaining particles. Hutchings and Lillford (1988) have described two thresholds that need to be satisfied before swallowing is initiated: a food particle size threshold and a lubrication threshold. Finally, debris can be left in the mouth after swallowing, and further clearance and swallows may be necessary.

18.2.3 Mechanisms of texture perception

Either of the two mechanisms described in Section 18.1.1 (proprioception and kinesthesis) can operate during the mastication process, depending on the nature and texture of the food. The texture of solid foods is perceived primarily through proprioception, as the food is chopped by the incisors and ground by the molars and through somesthesis as the bolus is formed and manipulated. The texture of semisolid and liquid foods is perceived primarily through somesthesis, and is usually expressed by the term ‘mouthfeel’.

18.3 Sensory assessment of texture

18.3.1 Principles of sensory assessment

A sensory stimulus to a human subject produces a set of physiological sensations that are interpreted by the brain as perceptions. The sensations will vary considerably between subjects, reflecting the natural physiological differences in the population, and the interpretation by the brain will be modified by psychological differences. Perceptions are recorded as actions by the subject, or as verbal responses. This may be the selection of a certain sample from a group (as in a difference test), words to describe the nature of a perception, and numbers to measure the size of a perception. To ensure that the subjects' responses relate as closely to their perception as possible, it is necessary to use carefully controlled environmental conditions and test procedures to carry out experiments. In particular, it is important to minimise the many sources of psychological bias that can produce unwanted influences on responses, and it is necessary to minimise the spread of physiological responses characteristic of biological systems through careful panel selection and training procedures (see also Chapter 2).

18.3.2 Quantitative procedures

A basic classification of the main sensory test procedures is shown in Fig. 18.2. The primary classification is into analytical tests and hedonic tests. Analytical tests use trained panels as a form of analytical instrument to generate information on the sensory properties of the food, whereas hedonic tests measure the response of untrained consumers to the sensory properties in terms of liking or acceptability. Different psychological procedures form the basis for this test distinction, and the information produced is distinct but complementary.

The most informative analytical procedures identify the sensory properties that are characteristic of that food, and quantify the individual characteristics; these are termed sensory profile methods. Numerous sensory profile methods have been developed for foods, but the most important in practical use are the Texture Profile Method, Quantitative Descriptive Analysis and the Spectrum Method (reviewed in Kilcast, 1999).

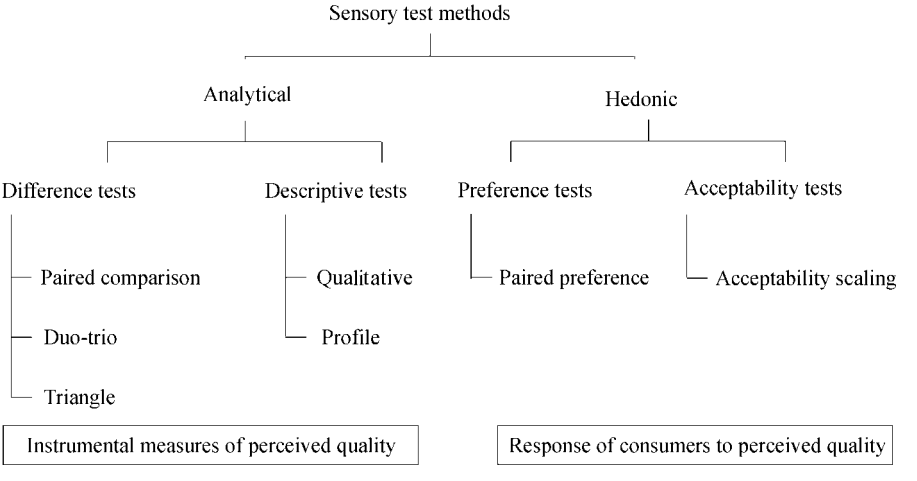


Fig. 18.2 Classification of the main sensory testing procedures.

The Texture Profile Method

The Texture Profile Method was developed by the General Foods Company specifically to define and measure the textural parameters of foods. Panellists are selected on the basis of ability to discriminate known textural differences in the specific product application for which the panel is to be trained (solid foods, beverages, semisolid foods, skin care products, fabrics and paper goods). Panellists selected for training are exposed to a wide range of products from the category under investigation, to provide a wide frame of references. The characteristics of the product, the order of appearance and the degree to which each is present are determined. Attributes usually are evaluated in the following order:

1. surface characteristics (can be visual)
2. initial compression (perceived on first bite)
3. masticatory phase (perceived during chewing)
4. residual phase (changes made during mastication and often perceived after swallowing).

In addition to the mechanical (e.g. firmness, adhesiveness, viscosity, springiness, cohesiveness) and geometrical (e.g. flakiness, grittiness, beady, crystalline) characteristics evaluated during the initial compression and masticatory stages, auditory characteristics such as crunchiness, crackliness or crispness might be evaluated. The panel verdicts are derived by group consensus or by statistical analysis of the data. Results are displayed in tabular or graphic form.

Quantitative Descriptive Analysis (QDA)

QDA is a total system covering sample selection, panellist screening, vocabulary development, testing and data analysis (Meilgaard *et al*., 1999). Variants of the original QDA procedures are probably used more than any other profiling procedure. The QDA technique uses small numbers of highly trained panellists. Typically, 6 to 12 people are screened for sensory acuity and trained to perform the descriptive task to evaluate the product. Three major steps are required: development of a standardised vocabulary, quantification of selected sensory characteristics and analysis of the results by parametric statistics.

Development of the vocabulary is a group process for creating a complete list of descriptors for the products under study. Panellists freely describe the flavour, appearance, odour, mouthfeel, texture and aftertaste characteristics of different samples. No hedonic (‘good’ or ‘balanced’), general (‘full’ or ‘typical’) or intensity-based (‘strong’ or ‘weak’) terms are permitted. Terminology should be consistent from product to product and tied to reference materials. The references decrease panellist variability, reduce the amount of time necessary to train sensory panellists, and allow calibration of the panel in the use of intensity scales. References should be simple, reproducible and clear to the assessors, and illustrate only a single sensory descriptor. They can be single chemical substances or finished products, and are made available during both the training and the testing phase, at various concentrations or intensity.

The attributes are collected and compiled into a master list. This individual preliminary evaluation of the samples may be revised during an open discussion to eliminate any redundant or synonymous descriptors. New terms might be added and physical references proposed. The panel leader condenses and formats the information into a proposal for standardised vocabulary. This vocabulary is then modified and improved in several interactive sessions. Multivariate statistical methods (e.g. factor

analysis) are sometimes used to reduce the number of descriptors. Finally, definitions for the attributes are agreed.

Once panellists feel comfortable with the vocabulary, further training is performed. The number of training sessions is dependent on the subject's performance, product and attribute difficulties and the time allowed for QDA testing. Panel training increases panellist sensitivity and memory and helps panellists to make valid, reliable judgements independent of personal preferences. Once the training sessions have established satisfactory panel performance, and removal of ambiguities and misunderstandings, the test samples can be evaluated. This is usually carried out in replicated (commonly three) sessions, using experimental designs that minimise biases.

In each session, the mean is calculated for group and individual judgements of each attribute. The results are then subjected to univariate statistics (e.g. multi-way analysis of variance) or multivariate statistics (e.g. principal component analysis). Test results may be also visualised via bar charts or line graphs. QDA diagrams must depict the relationships between products in a form that is readily understood with minimum of discussion.

The Spectrum Method

This more recent method provides a tool with which to design a descriptive procedure for a given product category. The method resembles QDA in many respects; for example, the panel must be trained fully to define all product sensory attributes, to rate the intensity of each and to include other relevant characterising aspects such as change over time, difference in the order of appearance of attributes, and integrated total aroma and/or flavour impact.

Panellists develop their lists of descriptors by first evaluating a broad array of products that define the product category. The process includes using references to determine the best choice of term and to best define that term so that it is understood in the same way by all panellists. Words such as vanilla, chocolate or orange must describe an authentic vanilla, chocolate, and orange character, for which clear references are supplied. All terms from all panellists are then compiled into a list that is comprehensive yet not overlapping.

The Spectrum Method is based on an extensive use of reference points. The choice of scaling technique may depend on the available facilities for computer manipulation of data and on the need for sophisticated data analysis. Whatever the scale chosen, it must have at least two, preferably three or five, reference points distributed across the range

18.3.3 Time-intensity methods

Sensory attributes are not perceived instantaneously, and can change in intensity with time in the mouth. Time-intensity methods are used to measure intensity of a specific attribute as a function of time in the mouth, and have been used to investigate the temporal behaviour of tastants, such as sweet and bitter molecules, and more recently have been used to investigate the release of volatile flavour materials from foods (Overbosch *et al*., 1991; Shamil *et al.*, 1992) during mastication. Such studies are particularly important in the reformulation of foods that results in structural modifications, and in changes that can occur on storage. These structural modifications are often accompanied by textural changes, and these often result in complex perceptual phenomena that are direct consequences of the changes in texture with time producing different flavour release phenomena. Although the use of time-intensity for flavour measurement is relatively well established, textural changes can also be monitored using the method (Burger, 1992; Duizer *et al*., 1993).

A major limitation of the time-intensity method is that only a single attribute can be tracked with time, and, if a number of important attributes are thought to be time-dependent, separate sessions are needed for each attribute. Difficulties encountered in time-intensity profiling prompted the development of a hybrid technique, progressive profiling (Jack *et al.*, 1994). In this technique, assessors carried out a profile on a set of texture descriptors at each chew stroke over the mastication period. Such a method has a number of potential advantages: several attributes can be assessed in one session; scaling is reduced to a unidimensional process; and the most important aspects of the shape of a time-intensity curve are retained.

18.4 Instrumental measurement of texture

Sensory methods are, for the foreseeable future, the primary means of measuring the range of textural characteristics of food that are important to consumer acceptance. The highly labour-intensive nature of sensory analysis has inevitably led to the development of instrumental methods designed to measure food properties that relate to relevant sensory characteristics. These methods have been classified in various ways, according to the type of measurement and the type of food.

18.4.1 Empirical, imitative and fundamental measurement

Instrumental methods have been classified into three main categories: empirical, imitative and fundamental.

Empirical methods

Empirical tests often measure ill-defined variables that are indicated by practical experience to be related to some aspect of textural quality. Devices have been developed within different sectors of the industry that are appropriate to specific product types. Even for the same product type, different food manufacturers have developed their own in-house devices. Fuller details of the devices described in this section are given in Bourne (1982).

Puncture or penetration devices measure either the force needed to push a probe into the food to a specified depth or the penetration distance achieved by application of a specified force. Examples include Magness-Taylor testers (for fruit), the Bloom Gelometer and the FIRA Jelly Tester (for gels), the cone penetrometer (for fats) and the Christel Texture Meter (for peas). Shearing devices measure the force needed for one or more blades to shear through the food. The maximum force is often assumed to measure toughness, firmness or fibrousness. Instruments include the Warner-Bratzler Shear (for meat), the Kramer Shear Cell (general-purpose) and the FMC Pea Tenderometer. Compression devices measure the force needed to achieve a given compression or the compression achieved at a given force. Examples include the Baker Compressimeter (for bread) and the ball indenter (for fats).

In extrusion tests, the food is forced through one or more orifices and the maximum force, average force or work done over a specified period is measured. The measured values are assumed to relate to firmness, toughness, consistency or spreadability. Examples include the FIRA-NIRD Extruder (for fats) and various cells used in conjunction with general-purpose instruments. Cutting devices use wires or blades (sometimes rotating) to cut through the food and measure the maximum force developed

or the time needed to cut through a standard size of sample. Measurements are assumed to relate to fibrousness, firmness or hardness. The FMBRA Biscuit Texture Meter is a rotating blade device used to measure biscuit hardness. Flow and mixing devices are used to give a measure of viscosity or consistency of liquid and semi-liquid foods. They often measure the extent to which samples flow or spread under specific geometric conditions, e.g. the Bostwick Consistometer and the Lyons Gel Flow Meter.

Although such empirical devices are often simple, inexpensive and portable, precision and reproducibility are generally poor, and the measured parameters are poor measures of perceived texture. Extensive use is still made of them in industry, however, mainly for quality control purposes.

Imitative methods

Imitative methods of measurement mimic the conditions to which the material is subjected in practice during eating. Volodkevich (1938) described the bite tenderometer, which attempted to mimic the action of teeth on food. It recorded the force of biting on a piece of food as a function of the deformation incurred. Two wedges with rounded points substituted for teeth, the lower being fixed to a frame. The upper wedge moved with a linear motion through the arc of a circle by a lever, squeezing a sample between the wedges.

The closest analogy to human chewing was attained by the denture tenderometer (Proctor *et al.*, 1955), an adaptation of Volodkevich's apparatus utilising a pair of human dentures to enable determination of food-crushing forces. The driving mechanism was able to impart both lateral and forward motion to the lower jaw. The amplified output from strain gauges placed on the articular arm measured the forces encountered. This served as the prototype for the General Foods (GF) Texturometer (Friedman *et al.* 1963), in which the dentures are replaced by a plunger. The location of the sensing element was moved from the articular arm to the sample area to eliminate gravity forces, and the oscilloscope was replaced by a chart recorder, enabling easy and permanent recording of any chosen number of consecutive chews. In this device, the driving mechanism no longer imparts a combined lateral and forward motion to the lower jaw, although it still drives the plunger through the arc of a circle. The linear speed of travel of the plunger varies sinusoidally with time (Brennan *et al.*, 1975).

A schematic diagram of the GF Texturometer is shown in Fig. 18.3. As a result of the complex dynamics of the instrument, interpretation of the data was of a semi-empirical nature. The force-time output was analysed in terms of a number of measured parameters, and in the original studies by the General Foods researchers, these were related to specific sensory attributes using the Texture Profile Method using detailed regression modelling.

Although the GF Texturometer remains in use to a small extent in North America and in Japan, the general-purpose testing machines designed for use with foods exemplified by those made by Stable Micro Systems, Stevens, Lloyd and Instron, are commonly used in the food industry in most countries. The instruments differ in their mechanical construction and in their data acquisition and data analysis capabilities, but they have a number of important features in common. All have a crosshead containing a load cell, which is driven vertically at a range of constant speeds, and which can cycle over a fixed distance or load range. Probes can be attached to the crosshead for penetrating, shearing or crushing food, which can be held in a variety of cells. The load is recorded relative to time or to penetration/deformation distance, and displayed on a suitable recorder. Computer control of the instrument and sophisticated and rapid computer analysis of the data are increasingly common. A typical modern instrument is shown in Fig. 18.4 together with examples of cells for specific food applications.

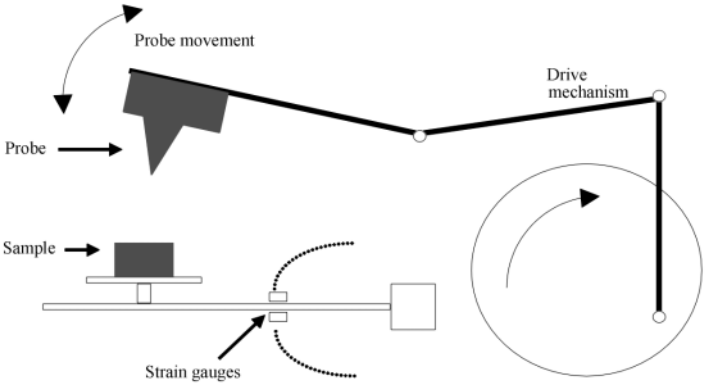


Fig. 18.3 Schematic diagram of the General Foods Texturometer.

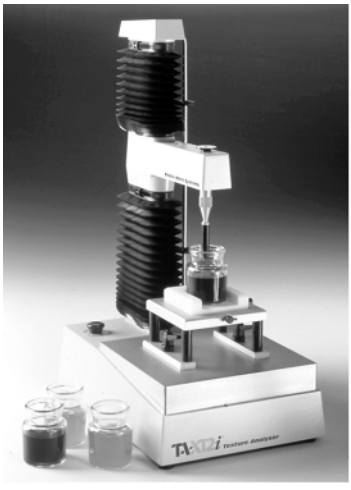
An idealised force-deformation relationship is shown in Fig. 18.5 together with some interpretations that are commonly placed on the measured parameters. It should be noted, however, that any interpretation should only be made via correlation of the instrumental parameters with quantified sensory attributes. This is too seldom the case in published literature. Force-time curves from real foods are in general considerably more complex than the above. Figure 18.6 shows the different curve shapes that result from testing a plain gelatin gel penetrated using probes of different geometries. Measurements of such curves require considerable care in both execution and interpretation.

A major advantage of such instruments is that flexibility of design allows them to be used for a wide range of foods. This is particularly useful for companies that are handling or manufacturing a varied product range. Load cells can be changed to give a high level of accuracy for relatively soft foods through to very hard foods. Probes and sample holders can easily be changed to accommodate measurements on different product types. An additional advantage is that such instruments can often be adapted for fundamental texture measurement.

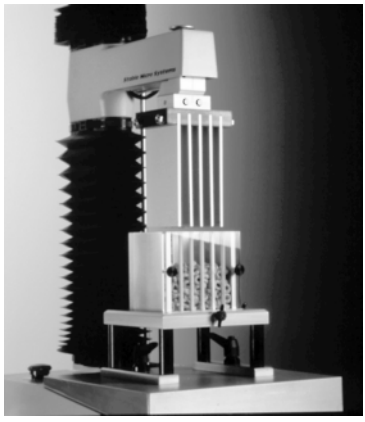
Fundamental methods

Fundamental methods involve measuring well-defined physical properties of food, which, if measured properly, are independent of the method of measurement. The most common fundamental parameters are Young's modulus, shear modulus, bulk modulus and Poisson's ratio (for solids) and viscosity (for liquids). Fundamental parameters for solids can be measured on general-purpose testing machines, but such measurements require a carefully designed experimental set-up and are consequently slow. In addition, foods are generally heterogeneous and do not exhibit simple elastic behaviour. Fundamental parameters can be measured on liquids using suitable instrumentation, for example the Weissenberg Rheogoniometer and the Carri-Med Rheometer. Again, however, liquid foods rarely exhibit simple viscosity behaviour.

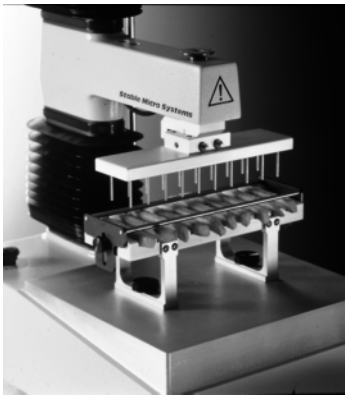
Such fundamental measurements are valuable in investigating the physical properties of food, but are too complex for routine use and, in common with other instrumental measurements, rarely correlate well with perceived texture. Some reasons for this can be identified on examining some of the physiological factors associated with chewing. In the following sections, only the principles underlying the measurement of the texture of solid foods will be covered; measurement of the texture of liquid foods using rheological principles will be described in Chapter 15.



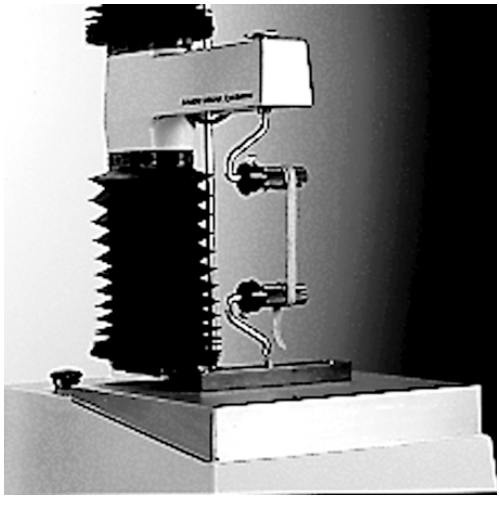
(a)



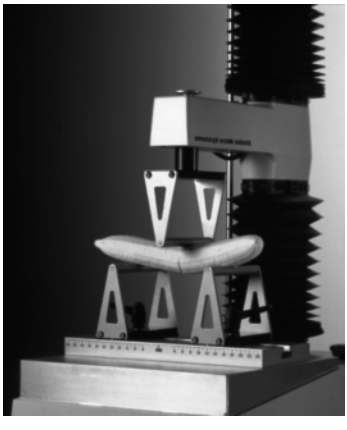
(b)



(c)



(d)



(e)

Fig. 18.4 An example of a modern general-purpose testing instrument: (a) general view of the instrument; (b) Kramer shear cell; (c) multiple chip rig; (d) spaghetti/noodle tensile rig; (e) three-point bend rig.

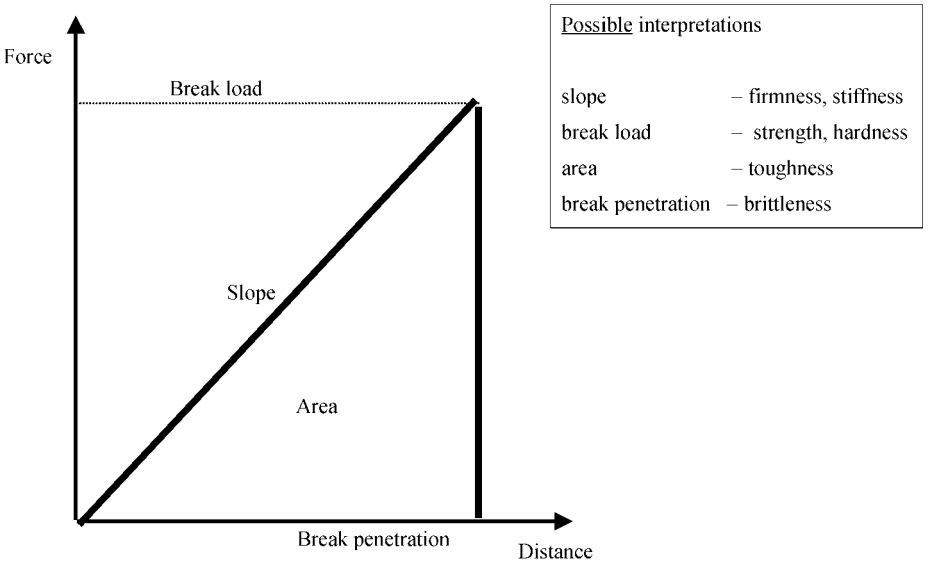


Fig. 18.5 Idealised force–deformation relationship.

18.4.2 Application to solid foods

Development of measurement methods

For most solid foods, key sensory attributes can be defined that are known to be highly important in defining consumer acceptability, for example crispness in salad vegetables and snack foods, tenderness in meat, and snap in chocolate. The evolution of measurement methods has followed the need to control the level of these attributes in routine production, but also the need to understand these attributes in designing new products.

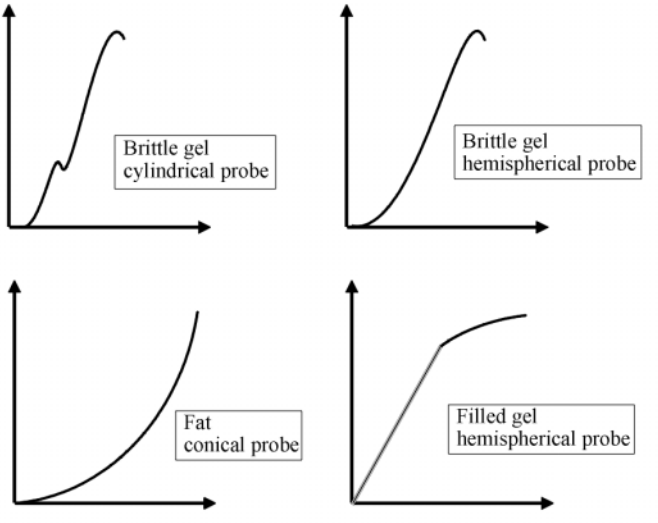


Fig. 18.6 Force–deformation curves recorded on testing different materials with a range of probe geometries.

The computerised modern instruments that utilise force-deformation principles are used almost universally in research functions and, with particular success, in quality control functions. A good example of this is the use by the French company Isigny Sainte-Mère of Stable Micro Systems TA-XT2 Texture Analysers for on-line measurement of the firmness of Camembert cheeses to sort for different maturing conditions (Toursel, 1996). The reason for the successful applications in QC is easy to see. In such applications, it is more important to be able to detect changes in measurable parameters than to measure precisely specific textural parameters. A change in any measured parameter outside set control limits can act as a signal that some aspect of the food production cycle has drifted. Of course, this introduces the risk that actions could be taken on the basis of changes in parameters that have little importance to consumer acceptability, but the measurements would normally trigger sensory tests that would minimise any production holds. A more serious risk is that the measurement system would not identify textural changes that were unmeasurable by that system, and, unless routine sensory tests were also carried out, defective material could be supplied to consumers. For example, a standard texture instrument capable of measuring the characteristic snap of chocolate might not detect the dryness that characterises stale chocolate.

The deficiencies of texture measurement instruments become particularly apparent when they are used for R&D purposes. One practical problem is that no single instrument is likely to be able to measure the food properties that are detected by both the tactile and the deep senses. One route to address this problem is to investigate more closely those physical processes that give rise to the texture sensations. In a study of stickiness in confectionery products, Kilcast and Roberts (1998) explained the need to understand both the meaning of the word ‘stickiness’ by consumers, and in what context the phenomenon is perceived. and also the physical processes that contribute to the various phenomena commonly described as stickiness. They showed that the perception of stickiness can occur through combinations of both adhesive and cohesive failure (Fig. 18.7).

Most industrial problems are associated with cohesive failure, which leaves unwanted material behind on surfaces. Both product rheology and the surface energy of the surfaces can contribute to the observed sticking phenomena, and under critical conditions it is possible to minimise sticking by either changing the surfaces involved, or changing product composition or operating conditions. The research led to a test procedure adapted from that developed by Chen and Hoseney (1995) for measuring the stickiness of dough that can be used to study the stickiness of caramels over a wide range of conditions. Figure 18.8 shows the cell, which consists of a water-jacketed cylinder holding the caramel sample ate the required temperature. The screw piston is used to extrude the caramel through a perforated stainless steel cap to give a fresh surface. The instrumental measurements are carried out by lowering a cylindrical probe onto the caramel surface, and then withdrawing the probe and measuring the force-deformation characteristics. For solid foods giving the proprioceptive (deep) response, the principles of fracture mechanics have been applied widely, and are outlined in the following section.

Fracture mechanics in food texture measurements

Classical measurements of the behaviour of food have commonly been made under low stress/strain conditions that cannot be expected to relate to the chaotic conditions prevalent in the mouth. Large deformation measurements have more practical significance, but many perceived textural properties depend on the way in which solid foods break down during mastication. In recent years, the principles of materials science that have successfully been used in the design of buildings, aircraft, bridges and automobiles have been applied to biological structures and to food (Vincent, 1992).

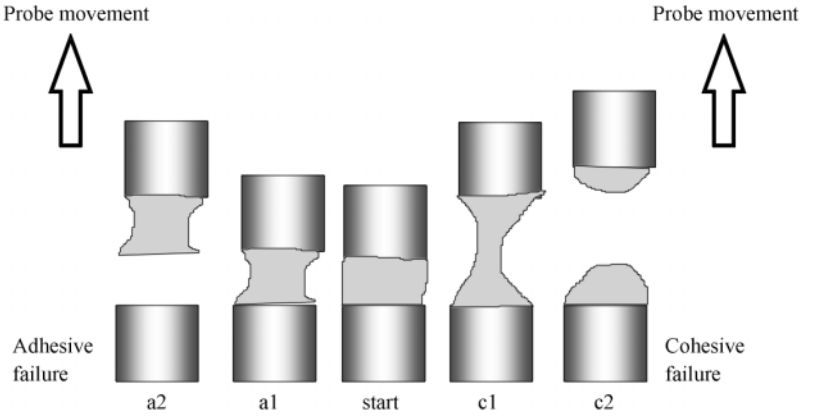


Fig. 18.7 Adhesive and cohesive failure mechanisms during the force–deformation testing of food. The sequence to the left (start, a1, a2) illustrates adhesive failure; the sequence to the right (start, c1, c2) illustrates cohesive failure.

The science of fracture mechanics was originally developed to explain the fracture of brittle materials such as glass. The breakage of a material is influenced by the relationship between the applied force and the bond holding the material together. Bonds will break more readily under the high local stress concentration around the tip of a sharp blade, for example. The basic premise underlying fracture mechanics is that all solids contain inhomogeneities, which exist in the form of flaws, or cracks. The magnitude and distribution of these defects govern the strength of the material. Fracture occurs when these defects grow and traverse the solid, creating new fracture surfaces. Early studies showed the potential of wedge fracture testing to brittle and semi-brittle foods such as apples and cheeses (Vincent *et al.*, 1991).

Two related criteria for the occurrence of fracture can be defined. The first is that fracture occurs when the stress intensity factor, K, at the crack tip exceeds a critical value, K_c, which is often referred to as the 'fracture toughness'. The magnitude of the stress intensity factor relates the stress intensity local to the crack tip in terms of the applied load and the geometry of the solid in which crack is located. Secondly, fracture can occur only if the stored strain energy of the system is equal to the energy required to generate new fracture surfaces. The critical strain energy release rate or, more simply, the fracture energy, G_c is the energy required to extending a crack over

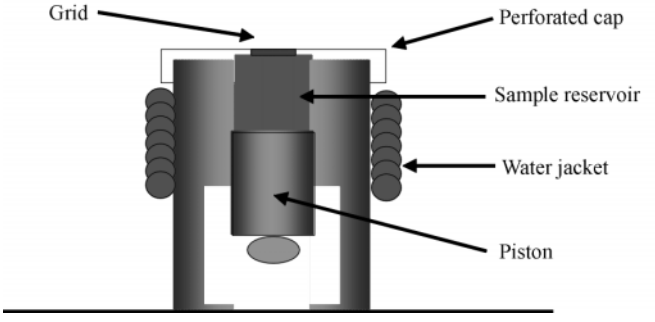


Fig. 18.8 Modified Chen-Hoseney cell for measuring the stickiness of caramel.

a unit area. Both the fracture energy, G_c and fracture toughness, K_c may be regarded as a measure of a material's resistance to crack propagation. The higher the value of G_c or K_c, the more resistant is the material to crack propagation (Kinloch and Young, 1983).

One difficulty in applying the principles of fracture mechanics to food lies in the great complexity of food structure, and in the complex viscoelastic behaviour of most foods, although recent research has demonstrated the feasibility of these techniques (Dolores Alvarez *et al*., 2000). However, deformation tests applied to carefully prepared food samples have been shown to give improved correlations between instrumentally measured properties and sensory measures. For example, measurements on apples, celery and carrots have show that a fracture toughness parameter gives good correlations with sensory hardness (Kilcast and Fillion, 2001). The principles of fracture mechanics have been extended to the measurement of fracture toughness of complex foods, such as the pastry casing of spring rolls (Sim *et al.,* 1993).

18.4.3 Analysis and validation of instrumental measurements

Statistical methods

A physical measurement of textural characteristics can be of practical value only if it is shown to relate to some relevant sensory texture measure. The relationship should take the form of a statistic that represents the fit between the instrumental measurement and the sensory attribute, or an equation that relates the instrumental measure (or set of measures) to the required sensory attribute. Two basic procedures are used:

- Pearson product moment correlation coefficients (r), where a perfect positive correlation gives $r = +1$, a perfect negative correlation gives $r = -1$, and no correlation gives $r = 0$. The square of the correlation coefficient (r^2) gives a measure of the data variance accounted for by the correlation, and is a measure of the value of the correlation.
- Multiple linear regression (MLR), in which the variable of interest (e.g. sensory attribute) is expressed as a linear combination of other variables (e.g. instrumental parameters). The variable combination is usually found using stepwise selection procedures. The method is of greatest value when the number of data points exceeds the number of attributes of interest.

The complexity of both instrumental and sensory data, however, increasingly demand the use of multivariate statistical procedures. Many techniques are available, but, when examining for structure and relationships in data sets, the most common technique is principal component analysis. PCA is a data reduction technique that replaces a large number of original variables by a smaller number of linear combinations, whilst still explaining a substantial proportion of the original variation in the data. Essentially, PCA projects an n-dimensional space onto a two-dimensional plot.

PCA analyses the correlation structure in the data set and identifies the axis along which the maximum variation occurs. A second principal axis is then identified orthogonal to the first axis, corresponding to the second greatest amount of variation, and so on. The new axes are linear combinations of the original axes, and the coefficients, or loadings, measure the importance of the original variables on each principal component. A useful reduction will often retain 70–80% of the variation in the first three dimensions. Plots of the loadings give valuable information on the relationships between variables, and especially on the correlation structure between attributes, and the possible use of

different terms for the same perceptual response. In addition, the locations of the samples can be plotted, and groupings identified in such a way that closely positioned samples are perceived as similar, and samples positioned far apart are perceived as very different. The samples plot is often superimposed onto the attribute plot to aid interpretation. It should be remembered, however, that, when analysing sensory data, useful information may be contained in minor dimensions that could be important in determining consumer response, and these dimensions, even if they contribute <10% to the total variance, should not be rejected out of hand. PCA is now a routine statistical procedure for analysing sensory profile data, and, in addition to the analysis of sensory profile data, PCA is increasingly used to examine for structure in combined sets of sensory and instrumental data. When analysing such data sets, it is important to carry out the analysis using the correlation matrix to compensate for the different measurement scales used. Other multivariate analyses, such as partial least squares analysis (PLS), are increasingly being used for combined data sets. For example, in a study of the sensory and instrumental characteristics of Reggiano grating cheeses, PLS was used to show that sensory texture correlated best with strain at breaking point (Hough *et al.,* 1996).

Novel methods for analysing force-deformation data

Analytical software included in the modern force-deformation test systems is capable of parameterising the curve shapes generated by many foods. The most extreme deviation from the idealised force-deformation curves shown in Fig. 18.9 is commonly found in testing brittle foods, which are characterised by very jagged curves. In a series of papers (e.g. Barrett *et al.,* 1992), Peleg and co-workers have described different mathematical approaches to analysing these highly irregular curve shapes, which can give rise to difficulties in parameterisation. One approach has been to carry out a Fast Fourier Transform on the force-deformation curve, giving a power spectrum of underlying frequencies. This procedure gives a qualitative representation of the jaggedness of the original curve, but cannot give a quantitative representation. Such quantification can be carried out through Fractal Analysis. The Fractal concept is based on the geometry of self-similar objects expressed in terms of non-integer dimensions. The fractal dimension is determined from the slope, in logarithmic co-ordinates between the length of the force-deformation contour and the corresponding measurement scale. The latter measurement was found to be convenient in giving a measure of overall jaggedness in terms of a single number, but the power spectrum gave more information on the location of the fracture, and its shape could be related more directly to structure and texture. The use of the power spectrum was subsequently described in relation to the measurement of crunchiness in friable baked products (Rohde *et al.,* 1993).

A further innovative attempt to analyse jagged force-deformation curves has been to use a technique used in speech analysis and medical diagnosis, symmetrised dot-pattern (SDP) displays (Peleg and Normand, 1992). This concept is based on the premise that, whereas humans find it difficult to analyse the visual appearance of highly irregular shapes, they are highly sensitive to changes in symmetric patterns. In the method, the recorded data are transformed into several symmetrically arranged sets of points, each reflected by a mirror plane, in an analogous way to the production of symmetric visual images by a kaleidoscope. The technique was used in the hope that the displays could be used to identify crunchiness in the same way that it can be used to identify vowels, but, in practice, the SDP displays were so sensitive to minor details that every signature appeared unique (Peleg, 1998).

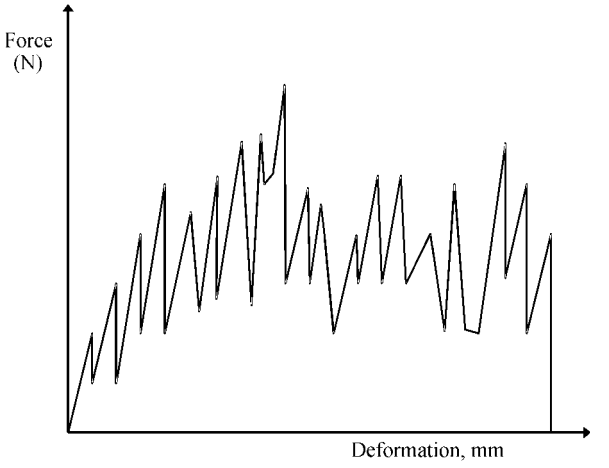


Fig. 18.9 Typical force–deformation curve generated on testing brittle foods such as dry snack foods.

18.5 *In vivo* texture measurement

The limitations in trying to mimic the events occurring during mastication using relatively simple instruments have long been appreciated. An alternative approach that has gained credence in recent years has been to attempt to record signals generated by or within the human subject that may relate to the texture of the food being masticated.

18.5.1 Electromyography (EMG) and associated techniques

EMG involves the use of a polygraph to measure electrical signals generated in muscles that are active during mastication. For certain muscles that lie close to the surface of the skin, for example the masseter muscle, which is active during the chewing of solid foods, this activity can be related to a specific muscle. Other oral activity, for example tongue movement, is controlled by groups of muscles that are deeper-lying. Monitoring of signals from this latter type of musculature ideally requires implanted electrodes, whereas signals from the masseter muscle can be readily recorded using surface electrodes.

Early attempts to use EMG in the study of food texture were limited by difficulty in interpreting the complex data patterns produced. In the absence of suitable computerised acquisition and analysis equipment, visual inspection of the raw data was carried out. For example, motor pauses (or silent periods) were more frequent with hard foods than with soft foods (Boyar and Kilcast, 1986). The development of more sophisticated EMG equipment and computer systems, however, has permitted much deeper analysis of EMG data and their relationship to food texture. An added potential advantage of this technique is studying changes in food texture in the mouth throughout the whole chewing cycle. Research projects carried out at the Leatherhead Food Research Association (Eves *et al.*, 1988; Eves, 1990; Kilcast and Eves, 1991; Eves and Kilcast, 1992), were carried out to investigate the potential use for EMG as a means of characterising food texture, and the way in which texture changes during mastication.

The raw EMG signal and its integrated form were recorded together with a time signal (Fig. 18.10). Fourier analysis of the raw signal was attempted, but proved inconclusive,

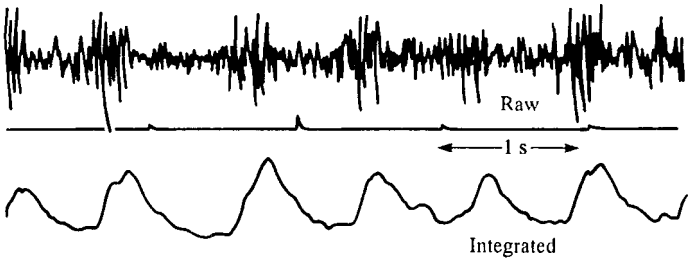


Fig. 18.10 Typical EMG output signals from the masseter muscle.

and the remainder of the studies used the integrated signal; measured parameters are shown in Fig. 18.11. Surface electrodes were used to detect the electrical signals from the masseter muscle. The skin on the cheek and earlobe was cleansed with 95% ethanol to remove traces of dust and perspiration, which may interfere with the signal. A position was located on the cheek at the maximum point of inflection of the masseter muscle, and two electrodes were located above and below this point in line with the muscle and approximately 0.5 cm apart. A third electrode was placed on the earlobe, a point of no muscular activity, which acted as an earth. Electrode cream, a conductive paste, was applied to the electrode surface in all cases.

Following connection of the terminal ends of the electrodes to the polygraph, the subject was presented with the sample to chew. The data acquisition system was capable of recording information from up to three muscles simultaneously; however, most experiments were carried out using a single masseter muscle, and consequently the subjects were asked to eat on the side of that muscle only. All samples were assessed in triplicate and all samples were of the same size and geometry. Data were recorded from the time the sample was put into the mouth to the time of swallowing.

Preliminary studies carried out to investigate subject-subject differences identified three groups characterised by differences in the way in which the integrated peak height (which was shown to be related to bite force, Fig. 18.12) varied with chewing time. Figure 18.13 shows the different curve types from the three groups chewing fruit pastilles. Seventy-five per cent of the subjects showed the group 2 behaviour, which

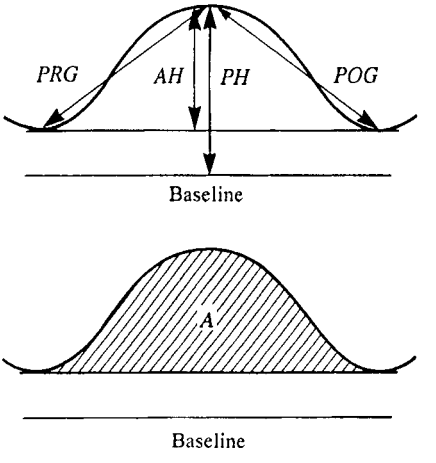


Fig. 18.11 Measured EMG parameters on the integrated signal. PH – peak height, AH – adjusted height, PRG – pre-maximum gradient, POG – post-maximum gradient, A – area.

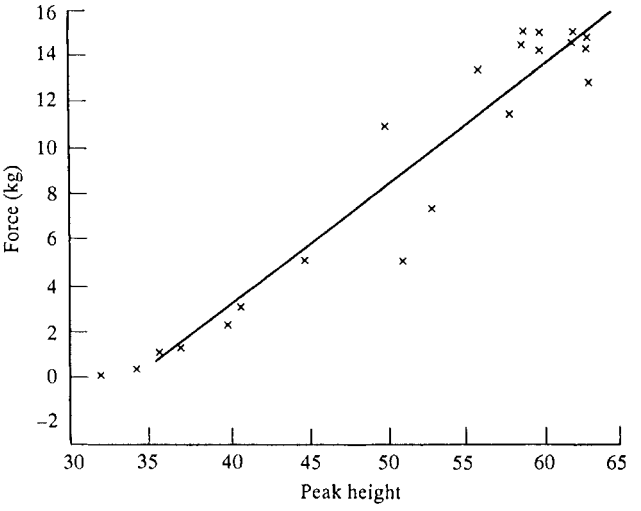


Fig. 18.12 Relationship between integrated peak height and measured bite force.

indicated a positioning and tentative biting phase before the main chewing phase. It was hypothesised that the group 1 curves characterised aggressive biting behaviour, and that the increasing biting force with time characteristic of group 3 was anomalous. Further studies were carried out using group 2 subjects, who also showed consistency in behaviour when used on different occasions.

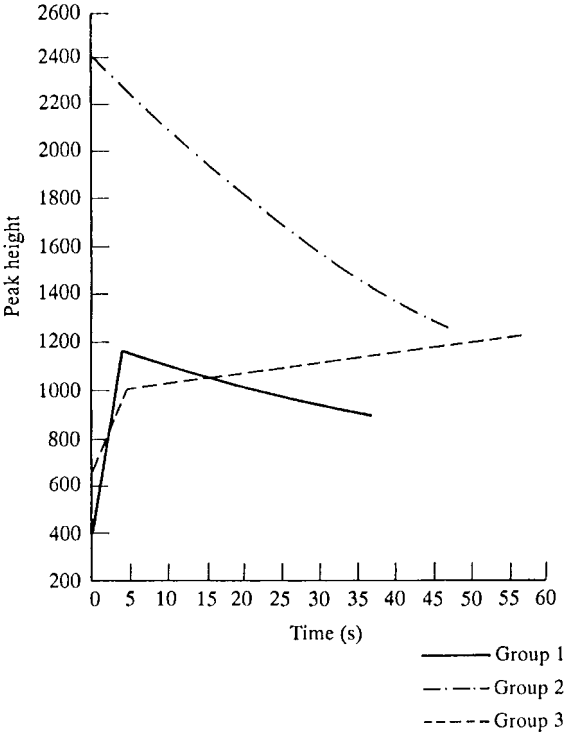


Fig. 18.13 Plots of integrated peak height against time, showing the behaviour of 26 subjects classified into three groups.

Subsequent investigations used confectionery products with a range of different textures. Exploratory data analysis highlighted the complex nature of the EMG data, and consequently multivariate methods were investigated. Multivariate methods in general seek to examine patterns in data points by finding linear combinations of the original experimental variables that account for most of the variations in the data. Canonical Variates Analysis (CVA) operates in a similar way, but defines groupings in the data by maximising the ratio of the variation between groups to the variation within groups (Kilcast, 1993). The first canonical variate is the direction that maximizes this ratio; the second is the orthogonal direction that maximises the remaining ratio; and so on.

Figure 18.14 shows the CVA plot from analysis of EMG parameters for the three commercial fruit pastilles. Each point represents one subject, and the dashed lines represent the 95% confidence limits for each grouping. The plot separates the three pastilles into three non-overlapping groups, with most subjects falling within the 95% confidence limits. Since CVA is designed to maximize the separation between groups, a check was carried out on the risk that the groupings might be an artefact of the method. Nine identical samples of wine gums were chewed, and the EMG data were treated as three different samples chewed in triplicate. The resultant CVA plots are shown in Fig. 18.15. The overlap of the 95% confidence circles shows that no distinct groupings were found in the data.

CVA plots on combined EMG data and sensory profile data also showed the potential for the EMG technique in product development. Figure 18.16 shows results from analysis of four types of model confectionery gums. The plot shows the wg50 sample to be well separated from the other three products, all of which overlap. The reason for this separation may be associated with the unique breakdown characteristics associated with this product, in which the wg50 sample exhibits rapid breakdown with no initial upward phase.

Following this work, a number of papers have appeared in the literature in which EMG has been used to investigate food texture. Further studies on the use of EMG in the confectionery sector were reported by Smalls (1992), which indicated good correlations between EMG and Instron measurements. Applications to cheese texture were described by Jack *et al.* (1993). In this work, EMG was used in conjunction with sensory and

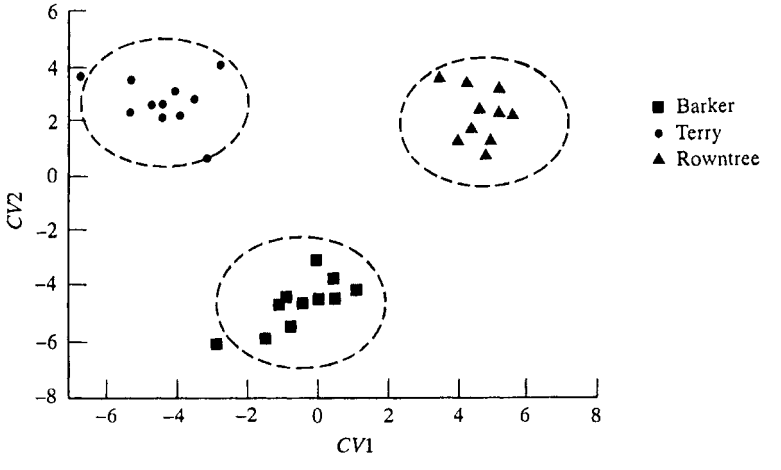


Fig. 18.14 CVA plots on EMG parameters measured from three commercial pastille confectionery gums.

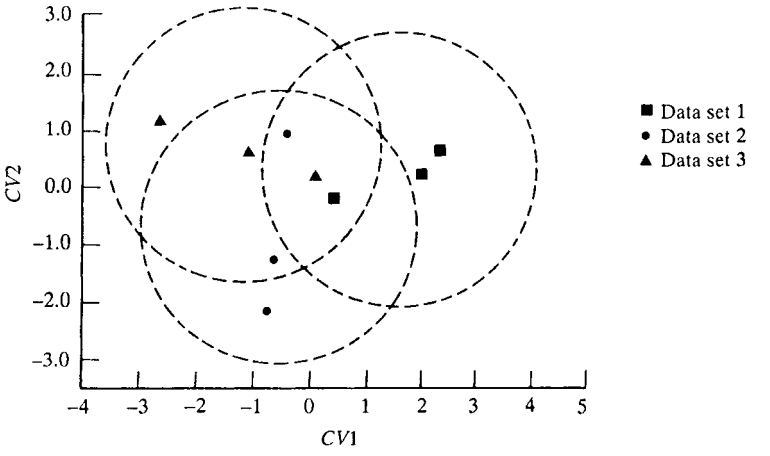


Fig. 18.15 CVA plots on EMG parameters measured from nine replicates of the same type of wine gum.

instrumental measurements of the texture of a range of Cheddar cheeses, but inconsistency between subjects resulted in difficulties in correlating EMG and Instron measurements. In a review of texture measurements for use in product development, Jack *et al.* (1995) described the use of EMG in conjunction with other methods. Duizer *et al.* (1996) used a combination of EMG, time-intensity measurement and instrumental measurement to investigate beef tenderness. The results indicated that the effects of early mastication should be compared with the effects of late mastication.

More fundamental aspects of the use of EMG in understanding the oral breakdown process have been reported by Brown and co-workers (e.g. Brown *et al.*, 1994; Brown, 1995; Brown *et al.*, 1998). These studies focused primarily on understanding the chewing behaviour of consumers rather than on texture measurement. EMG was used either as the sole technique, or in combination with synchronous measurement of jaw movement (kinesthesiology) by a set of transducers mounted on a head-frame to track the movement

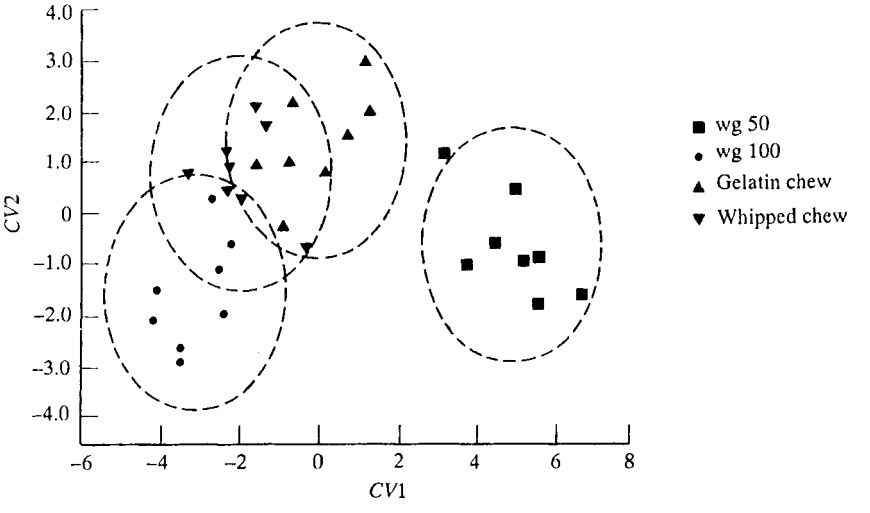


Fig. 18.16 CVA plots on EMG parameters measured from four variants of confectionery gums.

of a small magnet attached to the lower front incisors. Several oral techniques, including EMG, have also been used by Mioche and co-workers to study the mastication process (e.g. Mathevon *et al.*, 1995; Peyron *et al.*, 1996; Mioche and Martin, 1998).

An unusual related technique for studying mastication behaviour has been reported by Jack and Gibbon (1995). The technique, electropalatography (EPG), is used to measure tongue movement during eating and swallowing, and comprises an artificial plate, moulded to the individual's hard palate, embedded in which are 62 electrodes covering the entire palate surface. Tongue contact with the electrodes generates a signal that can be used to monitor the movement of the tongue. Experiments were carried out with liquid, semi-solid and gelled foods. The authors concluded that the technique could be used for liquid and semi-solid foods, but that bulky or sticky foods prevented the tongue making contact with the palate.

18.5.2 Sound emission

Early studies on food-crushing sounds (Drake, 1963) showed that sounds from crisp foods differed from those of non-crisp foods, primarily in terms of amplitude. Frequency and duration played a less important role. A subsequent paper (Vickers and Bourne, 1976) presented studies of the acoustical properties of tape-recorded biting sounds of wet and dry crisp foods. Amplitude-time plots indicated that both sound amplitude and the number of sounds produced in a given bite distance discriminated between different levels of crispness. A series of papers on crispness appeared subsequently (reviewed in Vickers, 1988). In one of this series (Vickers, 1985), it was shown that pitch was a useful parameter in distinguishing between crispness and crunchiness; crisp sounds tended to be higher in pitch than crunchy sounds.

Parts of this research programme examined the hypothesis that perceived crispness is a purely auditory sensation, but the work showed clearly that crispness has both auditory and non-auditory components. In a later study (Vickers, 1987), multiple linear regression was used to select combinations of mechanical and acoustical measurements that would best correlate with sensory crispness. For potato crisps, the following relationship was found:

$$\text{crispness} = -15.6 + 5.35NP + 133MHP - 6.21peak$$

where *NP* is the number of sounds during one bite, *MHP* is the mean height of peaks (both taken from oscilloscope displays of bite sounds) and *peak* is the peak of the force-deformation curve from a Kramer shear cell. For crisp breakfast cereals, the best relationship found was as follows:

$$\text{crispness} = 538 + 539(logMHP) - 222(peak)$$

These extensive studies gave considerable improvements to the understanding of crispness, and associated work has continued in other laboratories. Dacremont *et al.* (1992) investigated the contribution of both air-transmitted and bone-conducted sound to the perception of crispiness, crackliness and crunchiness in foods. In a further paper, Dacremont and Colas (1993) investigated whether a person's perception of duration, loudness and pitch of biting sounds might be biased by the sight of food samples. The results suggest that the visual perception of foods has no influence on the loudness or pitch judgement; however, it does influence the judgement of sound duration during eating. This paper pointed out that sound propagation in air is different between an

incisor bite, with lips open, and a molar chew, with lips closed. The contact area between food and teeth is different, and the amount of vibration through the bones is different. Although the bite sounds are different for the three textural types, the differences are not sufficient to distinguish them clearly. Roudaut *et al*. (1998) used acoustic emission in conjunction with sensory analysis, compression tests and Dynamic Mechanical Thermal Analysis (DMTA) to investigate the effect of water on the texture of crispy breads.

One difficulty in appraising the practical relevance of the research into sound emission and the texture of foods lies in the difficulty in the use of descriptive terminology. The word crispy and crunchy are, in particular, commonly used without agreed definitions. Recent work carried out at Leatherhead Food RA and Reading University on the measurement of the texture of fruit and vegetables has taken a consumer-led approach, and has investigated the understanding of textural terms by consumers. Repertory Grid interviewing methods were used to elicit textural descriptions of fruit and vegetables by consumers, and to relate them to attributes developed by trained sensory panels (Fillion and Kilcast, 2001). The trained panel developed a sensory profile that included different attributes associated with the sounds emitted during mastication.

Table 18.1 shows a comparison of term usage between consumers and panellists. Of the ten consumers interviewed, nine used the term 'crunchy' to describe the texture of the products, but only five mentioned the term 'crispy', while one did not mention 'crispy' or 'crunchy' at all. Those who mentioned both 'crispy' and 'crunchy' were always using the two terms with different meanings and usually found it easier to describe one against the other. There was generally a good agreement on the meaning of the term 'crunchy', with most definitions mentioning hardness and sound. Definitions for the 'crispy' attributes were more diverse, and were a combination of a snap clean break, a light texture and a sound, and were also associated with freshness, moistness and brittleness for some consumers.

These studies have assisted in developing improved methods for texture measurement. Measurement of sound can be used to show the close relationship between the release of

Table 18.1 Definitions of crispy and crunchy attributes from consumer and panellist interviews

	Consumers*		Panellists**	
	Crunchy	Crispy	Crunchy	Crispy
Number of subjects mentioning the attribute	9	5	11	10
Assessment on front teeth only		1		6
Assessment on back teeth only	1		6	2
Number of times a descriptor was given:				
Hard, dense	8		8	
Sound	8	3	10	9
Loud	2		4	4
Low pitch			8	
High pitch				8
Repetition of sound			5	
Brittle		1	3	
Snap, clean break		3		8
Light		2		6
Fresh		3		1
Moist		1		

* $n = 11$
** $n = 10$

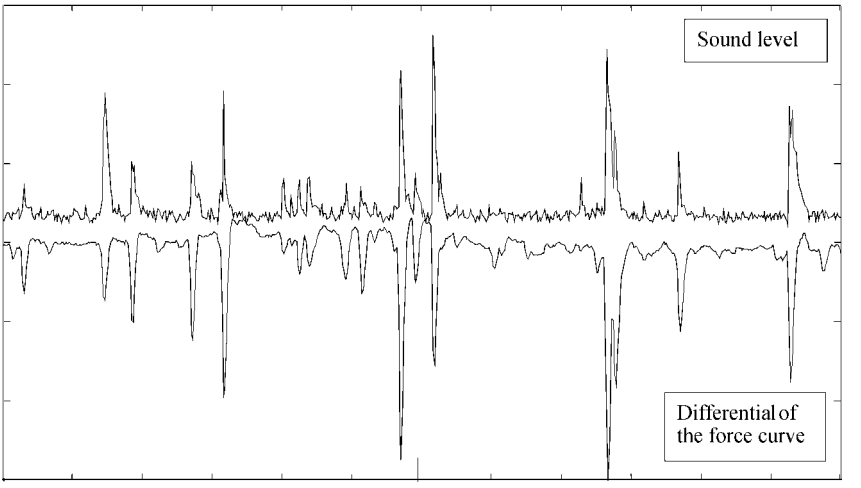


Fig. 18.17 Comparison of the emitted sound level with the differential of the force changes from cutting celery using an incisor probe.

sound energy and failure events that result in detectable changes in force. Figure 18.17 shows a comparison of force-deformation and emitted sounds for celery subjected to an incisor probe test in a Stable Micro Systems TA-HD instrument. The upper curve shows the sound emitted during the test, and the lower curve shows the differential of the force curve, measured from the signal taken directly from the load cell; the close correspondence is clear. However, substantial practical difficulties remain in the measurement of sound during the test, primarily from interferences by the noise emitted from the motor drive of the test machine, and noise from extraneous sources.

18.6 Future developments

Although it is increasingly being recognised that only sensory methods can give a complete picture of food texture, many needs can be identified for the instrumental measurement of key textural characteristics, and considerable research is being carried out to find new methods. There is particular interest in finding and validating non-destructive methods that can be used for quality control purposes, and for portable instruments that can be used directly in the field.

18.6.1 Dynamic force/deformation methods

The most common material parameters linked to firmness are the elastic properties of the produce: hard materials have high values of the elastic modulus, while soft materials have lower values (Pitts *et al*., 1994). Abbott and Massie (1995) believed that approximate values of the elastic properties of produce were adequate for sorting, and that a dynamic force/deformation (F/D) test might be relatively simple to implement on a kiwifruit packing line. They compared various parameters from a low-frequency F/D test with data from the Magness-Taylor puncture test, and found that a relatively simple firmness tester could be devised on the basis of 60-Hz vibrations.

18.6.2 Sound input methods

Sonic resonance testing is a potential method for non-destructive measurement of firmness. Resonant frequencies of intact fruits and vegetables decrease with ripening and are directly related to their rigidity, firmness and ripeness (Abbott *et al*., 1968). Abbott *et al*. (1995) showed that sonic stiffness coefficients were satisfactory predictors of the firmness of Golden Delicious apples, when compared with the maximum force in compression or puncture tests. Mealiness in apples has been studied using acoustic impulse methods (Tu and de Baerdemaeker, 1996). Chen *et al.* (1996) and Armstrong *et al*. (1997) also used the resonant frequency of the vibrational modes to successfully evaluate firmness of melons and peaches, respectively. Muramatsu *et al.* (1997) used pulsed sound transmission to measure the firmness of kiwifruit during ripening, and found good correlations with penetration testing. Low- frequency ultrasonics have been used (Nielsen and Martens, 1997) to evaluate the texture of carrots during cooking. Ultrasonic methods have also been used in the study of the texture of beef (Park *et al*., 1994), wafers (Juodeikiene *et al*., 1990) and crackers (Juodeikiene *et al*., 1994).

18.6.3 Spectroscopic and related methods

A number of both direct and indirect methods for assessing texture have been reported. Near-infra-red spectroscopy was investigated as a method for the quality control of asparagus (Garrido Varo *et al*., 2000). Laser scattering properties of apples were investigated as a means of measuring firmness (Cho and Han, 1999). Scatter parameters showed good correlations with compression testing, but strong varietal differences were found. Laser Doppler techniques have been used to measure firmness of fruit and to detect fruit disorders (Muramatsu *et al.*, 1999). The Doppler method gave good correlations with conventional instrumental testing on peaches, pears and citrus fruit, and is claimed to have potential as a remote-sensing device. Textural investigations using nuclear magnetic resonance (NMR) and magnetic resonance imaging (MRI) have also been reported. Schmidt (1999) has reviewed general uses of NMR methods, and Kim *et al*. (1998) have studied possible uses of both NMR and MRI for textural investigations.

18.6.4 Image analysis

Image analysis techniques have been reported for measuring different aspects of food texture. Affeldt *et al*. (1994). The paper reviews methods for the measurement of surface texture, dimension and shape and surface cracks. Coles and Wang (1997), described the use of a Bread Quality Imaging System to measure the fineness of bread texture by detecting bubbles above a minimum size. Measurement of the texture of corn puffs by image analysis has been described (Gao *et al.*, 1999). Visually determined sensory scores were effectively predicted by the intensity band image features. A further aspect of visualisation was demonstrated by Kilcast *et al.* (1984) and Boyar *et al.* (1984), in which the photoelastic properties of gelatin were used to visualise stress distributions in gelatin during the course of penetration testing. Figure 18.18 shows an example of the interference fringes produced on testing a gelatin gel with a hemispherically ended probe viewed under sodium light between crossed polarisers. Whilst such imaging is restricted to a small number of transparent gels, computation techniques are now available to calculate stress distributions in such systems, but do not appear to have been used in this context.

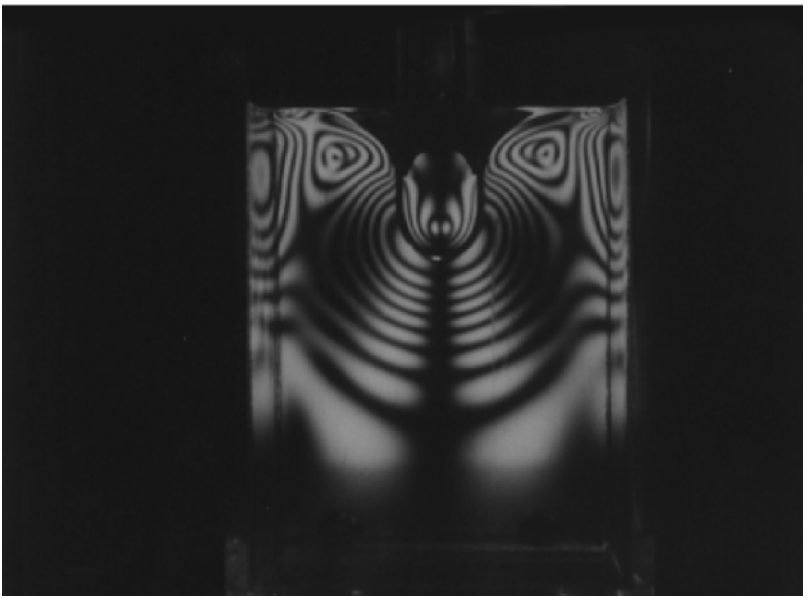


Fig. 18.18 Interference fringes produced on testing a gelatin gel with a hemispherically ended probe, and viewed under sodium light between crossed polarisers.

18.7 Conclusions

Given the complex nature of food texture, developments in measurement techniques are likely to be evolutionary rather than revolutionary. Can the ideal instrumental test, that incorporates the essential elements of fundamental tests, imitative tests and empirical tests be devised (Fig. 18.19)? That progress is being made is shown in Table 18.2, which summarises the status of texture measurement methods. Comparing this with the equivalent table in the first edition of this volume (Kilcast and Eves, 1993), important differences can be seen. More techniques are being investigated, and more are being used in production/QC environments. A number of factors have contributed to this situation. Firstly, conventional force-deformation instruments have undergone considerable enhancement through the extensive use of computerised data acquisition and analysis software, and through the availability of a wider range of test rigs for specific purposes. Secondly, the increased interest the mechanics of mastication has enhanced our understanding of oral processes and of their relationship to texture perception. Thirdly, the importance of texture as an important attribute influencing consumer acceptance has driven further research, particularly in the search for smaller but reliable measuring devices that can be used outside the QC laboratory. However, in recognition of the mportance of texture to the consumer, and in keeping with a food characteristic that is defined by the human senses, texture assessment is increasingly being carried out using trained sensory panels alongside the assessment of other organoleptic characteristics such as appearance, odour and taste.

Table 18.2 Summary of the main classes of texture measurement

Class	Food types	Initial costs	Running costs	Operating environment	Development status	Consumer relevance[6]
Sensory[1]	All	High	Moderate/high	Laboratory/QC/ production	Mature/continuing	High
Empirical[2]	All	Low	Low	QC/production	Mature/continuing	Low
Imitative[3]	Solids	Moderate	Low	Laboratory/QC	Continuing	Low/moderate
Fundamental[4]	All	High	Low	Laboratory	Mature	Low
Sound emission	Brittle solids	Moderate/high	Low/moderate	Laboratory	Continuing	Moderate/high
Sound input	Solids	Moderate/high	Low	Laboratory	Continuing	Low/moderate
Physiological[5]	Mainly solids	High/moderate	Moderate	Laboratory	Continuing	Moderate/high
Spectroscopic	Mainly solids	High	Moderate/high	Laboratory	Continuing	Moderate

[1] Trained profile panels
[2] e.g. hand-held penetrometers
[3] e.g. instrumental test rigs, General Foods Texturometer
[4] Force-deformation devices under strictly defined operational conditions
[5] Including EMG
[6] Consumer relevance depends on ability to correlate test with subjective textural responses

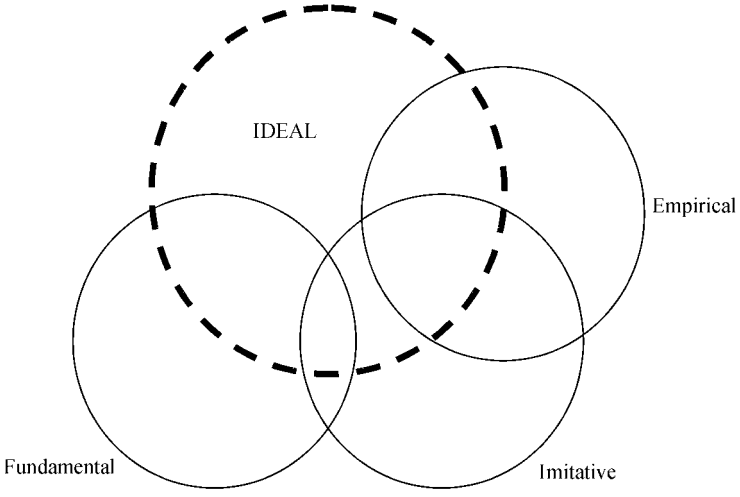


Fig. 18.19 Schematic illustration of the ideal instrumental test.

18.8 References

ABBOTT J.A., BACHMAN G.S., CHILDERS N.F., FITZGERALD J.V. and MATUSIK F.J. (1968). 'Sonic techniques for measuring texture of fruits and vegetables', *Food Technology*, **22(5)**, 101–12.

ABBOTT J.A. and MASSIE D.R. (1995). 'Nondestructive dynamic force/deformation measurement of kiwifruit firmness (*Actinidia deliciosa*)'. *Transactions of the ASAE*, **38(6)**, 1809–1812.

ABBOTT J.A., MASSIE D.R., UPCHURCH B.L. and HRUSCHKA W.R. (1995). 'Nondestructive sonic firmness measurement of apples'. *Transactions of the ASAE*, **38(5)**, 1461–6.

AFFELDT H.A., BROWN G.K., BRUSEWITZ G.H., DELWICHE M.J., HETZRONI A., KRANZLER G.A., PELEG K., SEARCY S.W. and SISTLER F.E. (1994). 'Dimension, shape and surface texture measurement on agricultural commodities'. In *Nondestructive technologies for quality evaluation of fruits and vegetables: proceedings of the international workshop, Spokane, Washington, June 1993*. American Society of Agricultural Engineers, Michigan, ASAE, pp. 50–62.

ARMSTRONG P.R., STONE M.L. and BRUSEWITZ G.H. (1997). 'Peach firmness determination using two different nondestructive vibrational sensing instruments'. *Transactions of the ASAE*, **40(3)**, 699–703.

BARRETT A.M., NORMAND M.D., PELEG M. and ROSS E. (1992). 'Characterization of the jagged stress-strain relationships of puffed extrudates using the Fast Fourier Transform and Fractal analysis'. *Journal of Food Science*, **57(1)**, 227–232, 235.

BOURNE M.C. (1982). *Food Texture and Viscosity*. Academic Press, New York.

BOYAR M.M. and KILCAST D. (1986). 'Food texture and dental science'. *J. Texture Studies*, **17**, 221–252.

BOYAR M.M., KILCAST D. and FRY J.C. (1984). 'Use of gel-based model food systems in texture measurement'. In *Gums and stabilisers for the food industry 2: Application of hydrocolloids*. Ed. G.O. Phillips. Pergamon Press, Oxford, 465–73.

BRENNAN J. G., JOWITT R. and WILLIAMS A. (1975). 'An analysis of the action of the General Foods Texturometer'. *Journal of Texture Studies*, **6**, 83–100.

BROWN W.E. (1995). 'The use of mastication analysis to examine the dynamics of oral breakdown of food contributing to perceived texture'. In *Characterization of Food: Emerging Methods*. Ed. A.G. Gaonkar, Elsevier, Amsterdam.

BROWN W.E., LANGLEY K.R., MARTIN A. and MACFIE H.J.M. (1994). 'Characterisation of patterns of chewing behaviour in human subjects and their influence on texture perception'. *Journal of Texture Studies*, **25**, 455–468.

BROWN W.E., EVES D., ELLISON M. and BRAXTON D. (1998). 'Use of combined electromyography and kinesthesiology during mastication to chart the oral breakdown of foodstuffs: relevance to measurement of food texture'. *Journal of Texture Studies*, **29**, 145–167.

BURGER J. (1992). 'Sensory evaluation techniques for chocolate with different types of cocoa butter products'. *Manufacturing Confectioner*, **72(10)**, 56–60.

CARDELLO A.V. (1994). 'Consumer expectations and their role in food acceptance'. In *Measurement of Food Preferences*. Eds. H.J.H MacFie and D.M.H. Thomson, Blackie A and P, London.

CHEN H., DE BAERDEMAEKER J. and BELLON V. (1996). 'Finite element study of the melon for nondestructive sensing of firmness'. *Transactions of the ASAE*, **39(3)**, 1057–65.

CHEN W.Z. and HOSENEY R.C. (1995). 'Development of an objective method for dough stickiness'. *Lebensmittel Wissenschaft und Technologie*, **28**, 467–73.

CHO Y.-L. and HAN Y.J. (1999). 'Nondestructive characterization of apple firmness by quantitation of laser scatter'. *Journal of Texture Studies*, **30(6)**, 625–638.

COLES G.D and WANG J. (1997). 'Objective determination of bread crumb visual texture by image analysis'. In *Proceedings of the International Wheat Quality Conference, Manhattan, May 1997*. Eds. J.L. Steele and O.K. Chung. Manhattan Grain Industry Alliance, pp 153–159.

DACREMONT C. and COLAS B. (1993). 'Effect of visual clues on evaluation of bite sounds in foodstuffs'. *Sciences des Aliments*, **13(4)**, 603–610.

DACREMONT C., COLAS B. and SAUVAGEOT F. (1992). 'Contribution of air- and bone-conduction to the creation of sounds perceived during sensory evaluation of foods'. *Journal of Texture Studies*, **22,** 443–456.

DOLOREZ ALVAREZ M., SAUNDERS D.E.J., VINCENT J.F.V. and JERONIMIDIS G. (2000). 'An engineering method to evaluate the crisp texture of fruit and vegetables'. *Journal of Texture Studies*, **31**, 457–473.

DRAKE B.K. (1963). 'Food crushing sounds: an introductory study'. *Journal of Food Science*, **28**, 233–241.

DUIZER L.M., GULLETT E.A. and FINDLAY C.J. (1993). 'Time-intensity methodology for beef tenderness perception'. *Journal of Food Science*, **58**, 943–947.

DUIZER L.M., GULLETT E.A. and FINDLAY C.J. (1996). 'The relationship between sensory time-intensity, physiological electromyography and instrumental texture profile analysis measurements of beef tenderness'. *Meat Science*, **42(2)**, 215–224.

EVES A. (1990). 'Physiological methods of food texture measurement: A thesis'. University of Reading.

EVES A., BOYAR M.M.. and KILCAST D. (1988). 'Electromyographic evaluation of the texture of confectionery products'. *Food acceptability: Proceedings of a symposium, University of Reading, September 1987*. Ed. D.M.H. Thomson, Elsevier Applied Science Publishers, London, pp 459–72.

EVES A. and KILCAST D. (1992). 'The texture of confectionery'. *Chem. Ind.*, **No 3**, 95–97.

FILLION L. and KILCAST D. (2001). 'Consumer perception of crispness and crunchiness in fruits and vegetables'. Accepted for publication in *Food Quality and Preference*.

FRIEDMAN H. H., WHITNEY H. and SZCZESNIAK A.S. (1963). 'The Texturometer – a new instrument for objective texture measurement'. *Journal of Food Science*, **28**, 390–403.

GAO X., TAN J., SHATADAL P. and HEYMANN H. (1999). 'Evaluating food properties by image analysis'. *Journal of Texture Studies*, **30(3)**, 291–304.

GARRIDO VARO A., SANCHEZ PINEDA DE LAS INFANTAS M.T. and PEREZ MARTIN M.D. (2000). 'A new analytical method applied to the quality control of asparagus: near infrared reflectance spectroscopy'. *Alimentaria*, June, 63–67.

HEATH M.R. and PRINZ J.F. (1999). 'Oral processing of foods and the sensory evaluation of texture'. In *Food Texture: Measurement and Perception*, Ed. A.J.Rosenthal, Aspen, Gaithersburg.

HOUGH G., CALIFANO A.N., BERTOLA N.C., BEVILLACQUA A.E., MARTINEZ E., VEGA M.J. and ZARITZKY N.E. (1996). 'Partial least squares correlations between sensory and instrumental measurements of flavor and texture for Reggiano grating cheese'. *Food Quality and Preference*, **7(1)**, 47–53.

HUTCHINGS J.B. and LILLFORD P. (1988). 'The perception of food texture: The philosophy of the breakdown path'. *Journal of Texture Studies*, **19**, 103–115.

International Standard ISO 5492 (1992); BSI 5098:1992. Glossary of terms relating to sensory analysis. ISO, Geneva.

JACK F.R., PIGGOTT J.R. and PATERSON A. (1993). 'Relationships between electromyography, sensory and instrumental measures of Cheddar cheese texture'. *Journal of Food Science*, **58(6)**, 1313–1317.

JACK,F.R., PIGGOTT J.R. and PATERSON A. (1994). 'Analysis of textural changes in hard cheese during mastication by progressive profiling'. *Journal of Food Science*, **59(3)**, 539—543.

JACK F.R. and GIBBON F. (1995). 'Electropalatography in the study of tongue movement during eating and swallowing (a novel procedure for measuring texture-related behaviour)'. *International Journal of Food Science and Technology*, **30 (4)**, 415–423.

JACK F.R., PATERSON A. and PIGGOTT J.R. (1995). 'Perceived texture: direct and indirect methods for use in product development'. *International Journal of Food Science and Technology*, **30(1)**, 1–12.

JUODEIKIENE G., PETRAUSKAS A., RIEDL O. and MOHR E. (1990). 'Applications of ultrasonics for the determination of wafer sheet texture'. *Ernahrung*, **14(9)**, 507–511.

JUODEIKIENE G., SCHLEINING G., KUNIGELIS V. and ADOMAITIS V. (1994). 'Theoretically and experimentally investigated correlation between acoustic and structural properties of crackers'. *Ernahrung*, **18(5)**, 235–237.

KILCAST D (1993). 'Analysis of textural data using canonical variates analysis'. In *Practical models for relationships between data sets: proceedings of the 3rd SENS plenary meeting*, Menorca, 1992. EC, Brussels.

KILCAST D. (1999). 'Sensory techniques to study food texture'. In *Food Texture: Measurement and Perception*. Ed. A.J.Rosenthal. Aspen, Gaithersburg.

KILCAST D. and EVES A. (1991). 'Integrating texture and physiology – techniques'. In *Feeding and the Texture of Food.* Eds. J.F.V. Vincent and P.J. Lillford. Society for Experimental Biology Seminar Series 44, Cambridge University Press, pp. 167–183.

KILCAST D. and EVES A. (1993). 'Modern methods of texture measurement'. In *Instrumentation and Sensors for the Food Industry*. Ed. E. Kress-Rogers. Butterworth-Heinemann, Oxford.

KILCAST D., BOYAR M.M. and FRY J.C. (1984). 'Gelation photoelasticity: a new technique for measuring stress distributions in gels during penetration testing'. *Journal of Food Science*, **49 (2)**, 654–5, 7.

KILCAST D. and FILLION F. (2001), 'Understanding consumer requirements for fruit and vegetable texture'. Accepted for publication in *Nutrition and Food Science*.

KILCAST D. and ROBERTS C. (1998). 'Perception and measurement of stickiness in sugar-rich foods'. *Journal of Texture Studies*, **29(1)**, 81–100.

KIM S.-M., MCCARTHY M.J., BIBBS D. and CHEN P. (1998). 'Water in tissue structures by NMR and MRI'. In *The Properties of Water in Foods: ISOPOW 6, Santa Rosa, 1996*. Ed. D.S. Reid. Blackie, London, pp. 30–39.

KINLOCH A.J. and YOUNG R.J. (1983). *Fracture Behaviour of Polymers.* Elsevier Applied Science Publishers, London.

MATHEVON E., MIOCHE L., BROWN W.E. and CULIOLI J. (1995). 'Texture analysis of beef cooked at various temperatures by mechanical measurements, sensory assessments and electromyography'. *Journal of Texture Studies*, 2**6(2)**, 175–192.

MEILGAARD M., CIVILLE G.V. and CARR B.T. (1999). *Sensory Evaluation Techniques* (3rd edn). CRC Press, Boca Raton.

MIOCHE L. and MARTIN J.-F. (1998). 'Training and sensory judgement effects on mastication as studied by electromyography'. *Journal of Food Science*, **63(1)**, 1–5.

MOSKOWITZ H.R. and KRIEGER B. (1995). 'The contributions of sensory liking to overall liking: An analysis of 6 food categories'. *Food Quality and Preference*, **6**, 83–90.

MURAMATSU N., SAKURAI N., YAMAMOTO D., NEVINS D.J., TAKAHARA T. and OGATA T. (1997). 'Comparison of a non-destructive acoustic method with an intrusive method for firmness measurement of kiwi fruit'. *Postharvest Biology and Technology*, **12(3)**, 221–228.

MURAMATSU N., SAKURAI N., WADA N., YAMAMOTO D., TAKAHARA T. and OGATA T. (1999). 'Evaluation of fruit tissue texture and internal disorders by laser Doppler detection'. *Postharvest Biology and Technology*, **15(1)**, 83–88.

NIELSON M. and MARTENS H.J. (1997). 'Low-frequency ultrasonics for texture measurement in cooked carrots (*Daucus carota L*)'. *Journal of Food Science*, **62(6)**, 1167–1170, 1175.

OVERBOSCH P., AFTEROF W.G.M. and HARING P.G.M. (1991). 'Flavour release in the mouth'. *Food Reviews International*, **7**, 137–184.

PARK B., WHITTAKER A.D., MILLER R.K. and HALE D.S. (1994). 'Ultrasonic spectral analysis for beef sensory attributes'. *Journal of Food Science*, **59(4)**, 697–701, 724.

PELEG M. (1998). 'Extracting useful information from irregular and irreproducible mechanical and other signatures'. In *New Techniques in the Analysis of Foods.* Ed. M.H. Tunick. Kluwer, New York.

PELEG M. and Normand M.D. (1992). 'Symmetrized dot-patterns (SDP) of irregular compressive stress-strain relationships'. *Journal of Texture Studies*, **4**, 427–438.

PAYRON M.-A., MIOCHE L., RENON P. and ABOUELKARAM S. (1996). 'Masticatory jaw movement recordings: a new method to investigate food texture'. *Food Quality and Preference*, **7(3–4)**, 229–237.

PITTS M.J., ABBOTT J.A., ARMSTRONG P.R., BROWN G.K., BRUSEWITZ G.H., DAVIS D.C., DELWICHE M.J., GALILI N., GAN-MOR S., HAUGH C.G., MASSIE D.R., MIZRACH A., NAHIR D., PELEG K., ROHRBACH R.P., SARIG Y., SCHAARE P.N., SCHNILOVITCH Z., SHMULEVICH I., STONE M.L., STROSHINE R.L. and YOUNCE F.L. (1994). 'Sensing fruit and vegetable firmness'. In: *Nondestructive technologies for quality evaluation of fruits and vegetables: Proceedings of the international workshop*, Spokane, Washington, June 1993 (American Society of Agricultural Engineers), pp 31–43.

PROCTOR B.E., DAVISON S., MALECKI G.J. and WELCH M. (1955). 'A recording strain gauge tenderometer for foods. 1: Instrument evaluation and initial tests'. *Food Technology*, **9**, 471–477.

ROHDE F., NORMAND M.D. and PELEG M. (1993). 'Characterization of the power spectrum of force-deformation relationships of crunchy foods'. *Journal of Texture Studies*, **24**, 45–62.

ROUDAUT G., DACREMONT C. and LE MESTE M. (1998). 'Influence of water on the crispness of cereal-based foods: acoustic, mechanical and sensory studies'. *Journal of Texture Studies*, **29**, 199–213.

SCHMIDT S.J. (1999). 'Probing the physical and sensory properties of food systems using NMR spectroscopy'. In *Advances in Magnetic Resonance in Food Science: Proceedings of the Fourth International Conference on Applications of Magnetic Resonance in Food Science, Norwich, September 1998*. Ed. P.S. Belton. RSC Cambridge, pp 79–94.

SHAMIL S.H., WYETH L.J. and KILCAST D. (1992). 'Flavour release and perception in reduced-fat foods'. *Food Quality and Preference*, **3(1)**, 51–60.

SHEPHERD R. and SPARKS P. (1994). 'Modelling food choice'. In *Measurement of Food Preferences*. Eds. H.J.H. MacFie and D.M.H. Thomson. Blackie A&P, London.

SIM B.J., LUCAS P.W., PEREIRA B.P. and OATES C.G. (1993). 'Mechanical and sensory assessment of the texture of refrigerator-stored spring roll pastry'. *Journal of Texture Studies*, **24**, 27–44.

SMALLS I. (1992). 'Electromyography versus Instron texture measurement of confectionery products'. *Proceedings of the 46th Annual Production Conference, Hershey, April 1992*. Pennsylvania Manufacturing Confectioners Association, Medford, pp 56–61.

SZCZESNIAK A.S. and KAHN E.L. (1971). 'Consumer attitudes to and awareness of food texture 1: adults'. *Journal of Texture Studies*, **2**, 280–295.

TOURSEL P. (1996). 'Equipment for texture analysis'. *Process*, **1122**, 48–50.

TU K., and DE BAERDEMAEKER J. (1996). 'Investigation of apple quality using instrumental methods'. *Agri-food Quality: an Interdisciplinary Approach; Proceedings of a Conference, Norwich, June 1995*. RSC Cambridge, 204–207.

VICKERS Z.M. (1985). 'The relationships of pitch, loudness and eating technique to judgements of the crispness and crunchiness of food sounds'. *Journal of Texture Studies*, **16**, 85–95.

VICKERS Z.M. (1987). 'Sensory acoustical and force-deformation measurement of potato chip crispness'. *Journal of Food Science*, **52**, 138–140.

VICKERS Z.M. (1988). 'Evaluation of crispness'. In *Food Structure – its Creation and Evaluation* Eds. J.M.V. Blanshard and J.R. Mitchell., Butterworths, Oxford.

VICKERS Z.M. (1991). 'Sound perceptions and food quality'. *Journal of Food Quality*, **14**(1), 87–96.

VICKERS Z.M. and BOURNE M C. (1976). 'A psychoacoustical theory of crispness'. *Journal of Food Science*, **41**, 1158–1164.

VINCENT J.F.V. (1992). *Biomechanics – Materials: A practical approach.* Oxford University Press, Oxford.

VINCENT J.F.V., JERONIMIDIS G., KHAN A.A. and LUYTEN H. (1991). 'The wedge fracture test. A new method for the measurement of food texture'. *Journal of Texture Studies*, **22,** 45–57.

VOLODKEVICH N.N. (1938). 'Apparatus for measurement of chewing resistance or tenderness of foodstuffs'. *Food Research*, **16**, 73–82.